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Explore Generative AI with the Gemini API in Vertex AI

Course · 5 hours ✓ Complete

Course overview

Explore Generative AI with the Gemini API in Vertex AI

Getting Started with Google Generative AI Using the Gen AI SDK

Multimodality with Gemini

Introduction to

Function Calling with Gemini

Explore Generative AI with the Gemini API in Vertex AI: Challenge Lab

Your Next Steps

Course Badge

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Lab setup instructions and requirements

End Lab 00:59:41

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-02-1f96235188860

Password

BPfNa0n3gr5Q

Project ID

qwik1abs-gcp-00-b9db72a1

Region

US-east1

Multimodality with Gemini

Lab 1 hour No cost Intermediate

★★★★★ Rate Lab

This lab may incorporate AI tools to support your learning.

GSP1210



Overview

This lab introduces you to [Gemini](#), a family of multimodal generative AI models developed by Google. You use the Gemini API to explore how Gemini Flash can understand and generate responses based on text, images, and video.

Gemini's multimodal capabilities enable it to:

- and compare visual similarities and differences.
- **Process videos:** Generate descriptions, extract tags and highlights, and answer questions about video content.

You experiment with these features through hands-on tasks using the Gemini API in Vertex AI.

Prerequisites

Before starting this lab, you should be familiar with:

- Basic Python programming.
- General API concepts.
- Running Python code in a Jupyter notebook on [Vertex AI Workbench](#).

Objectives

In this lab, you:

- Interact with the Gemini API in Vertex AI.
- Use the Gemini Flash model to analyze images and videos.

- responses.
- Explore practical applications of Gemini's multimodal capabilities.

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which prevents you from being able to reuse your account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

How to start your lab and sign in to the Google Cloud console

- Click the **Start Lab** button. If you need to pay for the lab, a dialog opens for you to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button
- Time remaining

Lab instructions and tasks

0/100

GSP1210

Overview

Objectives

Setup and requirements

Task 1. Open the notebook in Vertex AI Workbench

Task 2. Set up the notebook

Task 3. Use the Gemini Flash model

Congratulations!

- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in**

The lab spins up resources, and then opens another tab that shows the Sign in page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

student-02~1f9623518886@qwiklabs.net

You can also find the Username in the Lab Details pane.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

BPFNaOm3gr5Q

You can also find the Password in the Lab Details pane.

Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

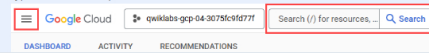
Note: Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To access Google Cloud products and services, click the **Navigation menu** or type the service or product name in the **Search** field.



Task 1. Open the notebook in Vertex AI Workbench

1. In the Google Cloud console, on the **Navigation menu** (☰), click **Vertex AI > Workbench**.
2. Find the **generative-ai-jupyterlab** instance and click on the **Open JupyterLab** button.

The JupyterLab interface for your Workbench instance opens in a new browser tab.

Note: If you do not see notebooks in JupyterLab, please follow these additional steps to reset the instance:

1. Close the browser tab for JupyterLab, and return to the Workbench home page.
2. Select the checkbox next to the instance name, and click **Reset**.
3. After the **Open JupyterLab** button is enabled again, wait one minute, and then click **Open JupyterLab**.

Task 2. Set up the notebook

1. Open the **intro_multimodal_use_cases** file.
2. In the **Select Kernel** dialog, choose **Python 3** from the list of available kernels.
3. Run through the **Getting Started** and the **Import libraries** sections of the notebook.
 - For **Project ID**, use **qwiklabs-gcp-00-b9db72ada886**, and for **Location**, use **us-east1**.

Note: You can skip any notebook cells that are noted *Colab only*. If you experience a 429 response from any of the notebook cell executions, wait 1 minute before running the cell again to proceed.

Task 3. Use the Gemini Flash model

Gemini Flash is a multimodal model that supports multimodal prompts. You can include

only images) and more in your prompt request and get text or video responses.

In this task, run through the specified notebook cells to see how to use the Gemini Flash model. Return here to check your progress as you complete the objectives.

Image understanding across multiple images

One of the Gemini's capabilities is being able to reason across multiple images. In this example, you use Gemini to calculate the total cost of groceries using an image of fruits and a price list.

Run through the **Image understanding across multiple images** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Image understanding across multiple images

Check my progress

Generating a video description

the video contents. In this example, you use Gemini to extract tags and retrieve extra information from different videos:

Run through the **Generating a video description** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Generating a video description

Check my progress

Audio understanding

Gemini can directly process audio for long-context understanding. In this example, you use Gemini to process audio for long-context understanding:

Run through the **Audio understanding** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Audio understanding

Check my progress

Reason across a codebase

Gemini can directly process audio for long-context understanding. In this example, you use Gemini to process audio for long-context understanding:

Run through the **Reason across a codebase** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Reason across a codebase

Check my progress

Video and audio understanding

In this example, you try out Gemini's native multimodal and long-context capabilities on video interleaving with audio inputs.:

Run through the **Video and audio understanding** section of the notebook.

☐

Video and audio understanding

Check my progress

All modalities (images, video, audio, text) at once

Gemini is natively multimodal and supports interleaving of data from different modalities. In this example, you try a mix of audio, visual, text, and code inputs in the same input sequence.

Run through the **All modalities (images, video, audio, text) at once** section of the notebook.

Click **Check my progress** to verify the objective.

☐

All modalities (images, video, audio, text) at once

Check my progress

Generating recommendations based on provided images

Gemini is capable of image comparison and providing recommendations. This is particularly useful for retail companies who want to provide users product recommendations based on their current setup.

Run through the **Generating recommendations based on provided images** section of

the notebook.

Click **Check my progress** to verify the objective.

☐

Generating recommendations based on provided images

Check my progress

Understand entity relationships in technical diagrams

Gemini has multimodal capabilities that enable it to understand diagrams and take actionable steps, such as optimization or code generation. In this example, you see how Gemini can decipher an entity relationship (ER) diagram, understand the relationships between tables, identify requirements for optimization in a specific environment like BigQuery, and even generate corresponding code.

Run through the **Understand entity relationships in technical diagrams** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Understand entity relationships in technical diagrams

Check my progress

Compare images for similarities and differences

Gemini can compare images and identify similarities or differences between objects. In this example, you use Gemini to compare two images of the same location and identify the differences between them.

Run through the **Compare images for similarities and differences** section of the notebook.

Click **Check my progress** to verify the objective.

☐

Compare images for similarities and differences

Check my progress

Congratulations!

You have now completed the lab! In this lab, you learned how to use the Gemini API in Vertex AI to generate text from text and image(s) prompts.

Next steps / learn more

Check out the following resources to learn more about Gemini:

- [Gemini Overview](#)
- [Generative AI on Vertex AI Documentation](#)
- [Generative AI on YouTube](#)
- Explore the Vertex AI [Cookbook](#) for a curated, searchable gallery of notebooks for Generative AI.
- Explore other notebooks and samples in the [Google Cloud Generative AI repository](#).

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

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